Homework: ESP32 → ThingSpeak

Goal

Send live temperature and humidity readings from an **ESP32** to **ThingSpeak**, and display the data publicly using built-in ThingSpeak charts.

What You Must Do

1. Hardware / Simulation Setup

Option A: Real Hardware

Use:

- ESP32 Dev Board
- DHT11 or DHT22 sensor

Wiring:

- VCC → **3.3V**
- GND → **GND**
- DATA → **GPIO 4** (or another pin—just mention it in your code)

Option B: Simulation (Recommended if no hardware)

Use **Wokwi**:

- ESP32 + DHT22
- Enable Internet simulation
- Use the Wokwi default pin (DATA → GPIO 15) or change as needed

2. Create a ThingSpeak Channel

Log in at https://thingspeak.com/

Create a new channel with the following name format:

FirstName_LastName_UniversityID
Example:
`Lina_Saleh_1203456`

Enable Public View so your graphs can be accessed via URL.

Add two fields:

- **Field 1:** Temperature (°C)
- **Field 2:** Humidity (%)

3. Program the ESP32

Use Arduino IDE, PlatformIO, or Wokwi's online editor.

Your program must:

1. Connect the ESP32 to Wi-Fi
2. Read temperature & humidity from DHT11/DHT22
3. Send the data to ThingSpeak every **15–30 seconds**
4. Include your **University ID** in the HTTP request as a parameter or in a printed log

Example (concept):
`https://api.thingspeak.com/update?api_key=XXXX&field1=23.5&field2=44.0`

4. Verify Cloud Data

You must show:

- **ThingSpeak channel page screenshot**
- **Public URL** (copy/paste)
- **Live temperature graph**
- **Live humidity graph**

Check that:

- Data is updating
- Graphs show a timeline
- Channel is publicly visible

Final Deliverables

Submit a **GitHub repository** containing:

A. Source Code
C. Screenshots
B. README.md with:
- Project description
- ThingSpeak public URL
- Wokwi link (if using simulator)
- Your full name + University ID
- Wiring diagram or Wokwi schematic